

Hands On - Data Analytics

Sorting, Grouping, Ranking & Visualization with Pandas

PYTHON

Code

```
import pandas as pd
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
# Create sample dataset
data = {
    'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhil', 'Fathima'],
    'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
    'Marks': [85, 78, 92, 88, 75, 95]
}
df = pd.DataFrame(data)
print("Original Data:")
print(df)
# 1. SORTING
print("\nSorted by Marks:")
sorted_df = df.sort_values(by='Marks', ascending=False)
print(sorted_df)
# 2. GROUPING
print("\nAverage Marks by Department:")
grouped_df = df.groupby('Department')['Marks'].mean()
print(grouped_df)
# 3. RANKING
df['Rank'] = df['Marks'].rank(ascending=False, method='dense')
print("\nStudents with Rank:")
print(df.sort_values(by='Rank'))
# 4. PLOTTING
# Bar chart of students' marks
plt.figure(figsize=(8, 5))
plt.bar(df['Name'], df['Marks'])
plt.title('Student Marks')
plt.xlabel('Student Name')
plt.ylabel('Marks')
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("plot1_student_marks.png", dpi=130)
plt.close()

# Plot average marks by department
grouped_df.plot(kind='bar')
plt.title('Average Marks by Department')
plt.xlabel('Department')
plt.ylabel('Average Marks')
plt.xticks(rotation=0)
plt.tight_layout()
plt.savefig("plot2_avg_marks_dept.png", dpi=130)
plt.close()
```

Output

Original Data:

	Name	Department	Marks
0	Arun	CSE	85
1	Bala	ECE	78

2	Chitra	CSE	92
3	Divya	ECE	88
4	Ezhil	CSE	75
5	Fathima	ECE	95

Sorted by Marks:

	Name	Department	Marks
5	Fathima	ECE	95
2	Chitra	CSE	92
3	Divya	ECE	88
0	Arun	CSE	85
1	Bala	ECE	78
4	Ezhil	CSE	75

Average Marks by Department:

Department

CSE 84.0

ECE 87.0

Name: Marks, dtype: float64

Students with Rank:

	Name	Department	Marks	Rank
5	Fathima	ECE	95	1.0
2	Chitra	CSE	92	2.0
3	Divya	ECE	88	3.0
0	Arun	CSE	85	4.0
1	Bala	ECE	78	5.0
4	Ezhil	CSE	75	6.0



